Velalar College of Engineering and Technology (Autonomous)

Empowering the Next Generation

THINDAL, ERODE - 638012

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25 Years of Academic Excellence 2001-2026

NAME	:	SAMSITHA S
REGISTER NUMBER	:	732924CSR135
COURSE CODE	:	22CST41
COURSE TITLE	:	DATABASE MANAGEMENT SYSTEM
ASSIGNMENT TITLE	:	ASSIGNMENT_3 - SQL QUERY PRACTICE

SAMPLE TABLE

Student

StudentID	Name	Department	Marks	City
201	Anu	CSE	85	Chennai
202	Bala	ECE	78	Coimbatore
203	Divya	CSE	92	NULL
204	Kiran	IT	74	Chennai
205	Rahul	ECE	88	Madurai

1. **Write an SQL query to display all student records from the Student table.**

```
CREATE TABLE Samsitha (
   StudentID INT PRIMARY KEY,
   Name VARCHAR(50),
   Department VARCHAR(10),
   Marks INT,
   City VARCHAR(50)
);
```

Output

SQL query successfully executed. However, the result set is empty.

```
INSERT INTO  Samsitha VALUES
(201,'Anu','CSE',85,'Chennai');

INSERT INTO Samsitha VALUES
(202,'Bala','ECE',78,'Coimbatore');
```

INSERT INTO Samsitha VALUES (203,'Divya','CSE',92,NULL);

INSERT INTO Samsitha VALUES (204,'Kiran','IT',74,'Chennai');

INSERT INTO Samsitha VALUES (205,'Rahul','ECE',88,'Madurai');

Output

SQL query successfully executed. However, the result set is empty.

SELECT * FROM Student;

Output

StudentID	Name	Department	Marks	City
201	Anu	CSE	85	Chennai
202	Bala	ECE	78	Coimbatore
203	Divya	CSE	92	
204	Kiran	IT	74	Chennai
205	Rahul	ECE	88	Madurai

2. **Write a query to display the Name and Marks of all students**.

SELECT Name, Marks FROM Samsitha;

Output

Name	Marks
Anu	85
Bala	78
Divya	92
Kiran	74
Rahul	88

3. Write a query to display students who scored more than 80 marks.

```
SELECT * FROM Samsitha
WHERE Marks > 80;
```

Output

StudentID	Name	Department	Marks	City
201	Anu	CSE	85	Chennai
203	Divya	CSE	92	
205	Rahul	ECE	88	Madurai

4. Write a query to display students belonging to the CSE department.

```
SELECT * FROM Samsitha
WHERE Department = 'CSE';
```

Output

StudentID	Name	Department	Marks	City
201	Anu	CSE	85	Chennai
203	Divya	CSE	92	

5. write a query to display students whose city is Chennai.

SELECT * FROM Samsitha WHERE City = 'Chennai';

Output

StudentID	Name	Department	Marks	City
201	Anu	CSE	85	Chennai
204	Kiran	IT	74	Chennai

6. Write a query to display students whose marks are between 70 and 90.

SELECT * FROM Samsitha
WHERE Marks BETWEEN 70 AND 90;

Output

StudentID	Name	Department	Marks	City
201	Anu	CSE	85	Chennai
202	Bala	ECE	78	Coimbatore
204	Kiran	IT	74	Chennai
205	Rahul	ECE	88	Madurai

7. **Write a query to display student names that end with the letter ‘a’ using the LIKE operator.**

```
SELECT * FROM Samsitha
WHERE Name LIKE '%a';
```

Output

Name
Bala
Divya

8. **Write a query to display distinct departments from the Student table.**

```
SELECT DISTINCT Department
FROM Samsitha;
```

Output

Department
CSE
ECE
IT

9. **Write a query to display students whose City value is NULL.**

```
SELECT * FROM Samsitha
WHERE City IS NULL;
```

Output

StudentID	Name	Department	Marks	City
203	Divya	CSE	92	

10. Write a query to find the highest marks obtained by a student using an aggregate function.

SELECT MAX(Marks) AS HighestMarks
FROM Samsitha;

Output

MAX(Marks)
92

11. Write a query to find the average marks of students in the ECE department.

SELECT AVG(Marks) AS AverageMarks
FROM Samsitha
WHERE Department = 'ECE';

bg

Output

AVG(Marks)
83

12. Write a query to count the number of students in each department using GROUP BY.

```
SELECT Department, COUNT(*) AS StudentCount
FROM  Samsitha
GROUP BY Department;
```

Output

Department	COUNT(*)
CSE	2
ECE	2
IT	1

13. Write a query to display students whose marks are greater than the average marks of all students (nested subquery).

```
SELECT * FROM Samsitha WHERE Marks > (
   SELECT AVG(Marks) FROM Samsitha);
```

Output

StudentID	Name	Department	Marks	City
201	Anu	CSE	85	Chennai
203	Divya	CSE	92	
205	Rahul	ECE	88	Madurai

14. Write an SQL statement to insert a new student record into the Student table.

```
INSERT INTO Samsitha VALUES
(206,'Meena','CSE',81,'Salem');
```

Output

SQL query successfully executed. However, the result set is empty.

15. Write an SQL query to delete the student with StudentID = 205.

DELETE FROM Samsitha WHERE StudentID = 205;

Output

SQL query successfully executed. However, the result set is empty.